Graph Representations for Congestion Onset Prediction: A Controlled Comparative Study

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Abstract. Graph models offer a natural representation of traffic networks, but their contribution to regional congestion warnings can be difficult to separate from the benefits of additional sensor data and updated training. We present a controlled comparison of six temporal and graph representations on PEMS-BAY, examining local sensor histories, graph structure and aggregate model updating within a common prediction framework. The study combines competitive aggregate baselines, trained randomized-graph controls and a held-out chronological test with 9,315 eligible prediction times. Aggregate refitting achieves the lowest observed prediction error, while the local-history correction performs worse than its aggregate counterpart. A graph model that separates within-region and cross-region connections outperforms both randomized controls in secondary comparisons and reduces Brier score by 1.11% relative to the aggregate correction. However, favorable primary graph comparisons remain statistically uncertain after adjustment for multiple comparisons. The cross-region model also raises the false-alarm rate from 5.97% to 7.12% without improving recall. These findings reveal how representation choice can affect probability accuracy and alarm quality differently. Our comparison provides a reproducible framework for assessing the added value of spatial modeling and shows why competitive aggregate updating and alarm evaluation are essential to interpreting graph-based congestion forecasts.

Keywords: Traffic forecasting · Congestion onset · Graph representations · Aggregate baselines · Chronological evaluation

1 Introduction

Predicting the onset of regional congestion is a central task for traffic monitoring and early warning. The objective is to estimate whether sustained slowdowns will

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soon affect a substantial part of a network. This requires a different prediction target from forecasting the speed at every sensor. A regional warning depends on the extent and persistence of congestion, making the choice of representation particularly important. Detailed sensor histories may reveal emerging spatial patterns, while regional summaries may already capture much of the information needed to predict the event.

Graph-based forecasting provides a natural way to organize traffic measurements through spatial relationships. Established approaches combine temporal modeling with diffusion operators or learned connectivity [5,11]. At the same time, competitive alternatives demonstrate the importance of strong baselines when assessing the contribution of graph processing [6,10]. For regional congestion warnings, the central question is what spatial detail and graph structure add beyond a well-trained aggregate predictor. Answering this question requires distinguishing the effects of additional measurements, the operations applied to them, and access to more recent training data.

These distinctions matter because an apparent graph advantage can have several explanations. A model may benefit from individual sensor histories without benefiting from their connectivity. It may improve because it has more parameters or was trained on additional observations. A reduction in probability error may also coincide with more false alarms. Evaluating these effects together provides a clearer basis for selecting a forecasting approach and for deciding whether its spatial processing is justified.

We address this problem through a controlled comparison on PEMS-BAY. Six compact representations cover local temporal histories, distance-based sensor messages, regional messages, multiscale graph summaries, separate within-region and cross-region messages, and training-estimated dependencies. Each representation enters a common prediction framework through an additive correction to a frozen aggregate probability model. This design keeps the prediction target and much of the fitting procedure consistent while varying the available detail and its processing.

The aggregate controls are central to the comparison. In particular, R_{union} refits the aggregate predictor on the union of the original and correction training examples. It therefore tests whether updating the aggregate model is competitive with adding local or graph representations. A larger aggregate correction assesses the role of parameter budget, while trained rewired graphs test whether the observed connectivity provides an advantage over alternative connections.

The study makes three empirical contributions:

1. A common comparison of six representations for regional congestion onset, supported by aggregate controls that account for additional training data and model capacity.
2. An assessment of topology-specific predictive evidence using trained graph controls, three predeclared correction seeds, and a chronological test whose models and analysis were fixed before access.

3. A joint evaluation of probability accuracy, calibration, alarm behavior and measured processing costs, showing how these criteria affect the interpretation of representation gains.

The results illustrate the value of this combined evaluation. Aggregate refitting achieves the lowest observed prediction error among the main model families, while adding local histories through the tested correction does not improve the aggregate baseline. Separating within-region and cross-region messages yields favorable secondary comparisons against trained rewired controls, but the favorable primary graph contrasts remain statistically uncertain and false alarms increase without a recall gain. These findings provide concrete guidance for regional traffic forecasting: assess spatial representations against competitive aggregate updating, test the contribution of connectivity explicitly, and judge predictive gains alongside the quality of the resulting warnings.

2 Related Work

Graph-based traffic forecasting combines temporal history with a spatial dependency model. DCRNN uses bidirectional diffusion and recurrent forecasting [5]. Graph WaveNet combines temporal convolutions with an adjacency learned through node embeddings [11]. These methods motivate testing both fixed and measurement-derived connectivity. Our distance and dependency arms are much smaller controlled variants and predict a regional event probability directly. Their results cannot rank the original architectures on their established speed-forecasting benchmarks.

The need for strong simple spatial baselines is already documented. *Do We Really Need Graph Neural Networks for Traffic Forecasting?* introduces SimST, which models local proximity and global correlations without conventional message passing [6]. STID uses spatial and temporal identity information with MLPs [10]. These studies show why comparing a graph model only with a weak non-spatial predictor is insufficient. Our local encoder retains regional identity and ordered histories, but is not a reproduction of SimST or STID. The additional question here is whether local and graph inputs help a rich aggregate event predictor after matching access to later labels.

Standardized evaluation is equally relevant. BasicTS+ examines inconsistent multivariate forecasting comparisons and supplies a common training pipeline [9]. LargeST provides a broader traffic-flow benchmark [7]. We use neither their full model roster nor their standard scoring task. We retain one existing speed dataset and an explicitly defined chronological risk endpoint. The graph benchmarking framework of Dwivedi et al. motivates parameter-budget controls and reproducible comparison [3]; comparable parameter counts do not make different representations equally expressive.

Residual correction also has direct traffic precedents. ResCAL estimates errors of existing forecasts from previous errors and graph signals [4]. Our corrections instead learn a bounded additive change to a frozen event probability from

histories and summaries. Residual learning itself is established, and this input/-target difference does not establish priority. We evaluate whether the mechanism is useful relative to aggregate updating.

Graph aggregation provides an analytical language for describing lost detail. Equitable partitions and quotient dynamics have established conditions under which grouped states close under an operator [8]. Diffusion wavelets develop multiscale representations of diffusion operators [1]. The decomposition used below is elementary projection algebra. We use it to define candidate summaries, without claiming a new coarsening theorem or sufficiency for nonlinear congestion events.

3 Task and Observation Model

3.1 Recorded-Speed Event and Onset Eligibility

PEMS-BAY contains five-minute speed measurements in mph for 325 ordered sensors from January through June 2017 [5,2]. Let $v_{i,t}$ denote recorded speed and f_i the fixed reference free speed for sensor i, estimated as its 85th percentile within the original training period. For fully observed frames, define

$$c_t = \sum_{i=1}^{325} \mathbf{1}\{v_{i,t} \le f_i/2\}, \qquad Y_t = \mathbf{1}\left\{ \max_{j=1,\dots,4} \min_{k=0,1,2} c_{t+j+k} \ge 48 \right\}. \tag{1}$$

Here, c_t counts sensors recording at most half their reference speed. The event $Y_t = 1$ requires at least 48 such sensors in each of three consecutive frames within the next six frames, a 30-minute forecast horizon. The predictor uses the preceding 12 frames, including t. Onset evaluation includes only forecast origins where the current three-frame event can be ruled out. It therefore concerns a future regional event, rather than the onset time at an individual sensor or an annotated road incident.

Nonfinite and nonpositive measurements are treated as unknown. For each future frame, a lower count excludes unknown sensors from the slow-speed count, while an upper count includes them. Applying Eq. 1 to both count sequences bounds the event label. An origin is scoreable only when these bounds agree and at least 95% of its future sensor-frame entries are observed. Missing future speeds are therefore not imputed to determine labels. For current onset eligibility, the upper counts exclude origins where missing observations could conceal an active event. Every model uses the same eligibility and scoring masks.

These labels describe sustained low recorded speeds. They do not identify a specific road incident, establish causal propagation along graph edges, or demonstrate that acting on an alarm would improve traffic conditions. Those interpretations require additional observations and operational evaluation.

3.2 Aggregate and Local Inputs

Sensors are assigned to 16 fixed regions through a deterministic recursive partition of their coordinates. The aggregate vector contains 1,158 entries. These com-

prise 12 frames of regional means, valid observation counts, variances, minima, maxima and severe fractions, together with six calendar features. The summaries capture variation within regions as well as average slowing. Their computation requires readings from every sensor, so aggregation does not reduce sensor acquisition requirements.

Input speeds are transformed using reference speeds estimated from an earlier training period,

$$x_{i,t} = \operatorname{logit}\Big(\operatorname{clip}_{[.001,.999]}(1 - v_{i,t}/f_i^{\text{feature}})\Big). \tag{2}$$

The reference speeds used for features and event labels have different training cutoffs. Missing transformed values are set to zero and accompanied by observation masks and valid counts. Aggregate correction inputs are standardized using the mean and scale estimated from correction training data, then clipped to $[-10, 10]$.

Local models receive each sensor's transformed history in temporal order, its deviation from the regional mean, its recent change when both frames are valid, and an observation validity channel. A temporal multilayer perceptron (MLP) shares parameters across sensors to encode these histories. Pooling with observation masks produces one vector per region, accompanied by regional availability fractions. These vectors remain in a fixed regional order, preserving regional identity even without graph messages. Local models therefore supplement the aggregate vector with individual sensor detail. Changing the graph while retaining the same histories changes how this information is combined across sensors.

4 Predictors and Graph Representations

4.1 Frozen Anchor and Aggregate Controls

The anchor A_G provides the baseline probability to which corrections are added. It is a GPU tree booster using quantile bins, with 100 trees, at most 15 leaves per tree, 64 candidate bins, a minimum of 20 examples per leaf, and learning rate .1. Its L2 regularization was selected from two settings during early training. R_{union} retains the selected model specification and feature transform but is refitted on the exact union of anchor and correction training examples, using the same sampling stride. Both fitted models are saved and produce separate predictions. All corrections use A_G as their anchor.

A_{mono} transforms the frozen anchor probabilities through a logistic map with positive slope. The map was fitted using anchor predictions from the correction period, outside the anchor's fitting data. B learns a compact correction from aggregate inputs. K uses a larger aggregate correction with approximately as many parameters as the local and graph corrections. Training selected B as the aggregate comparator and E as the expanded graph comparator. These selections remain fixed for test regardless of subsequent model rankings.

Every correction takes the form

$$p_t = \operatorname{clip}_{[0,1]}\{p_{A_G,t} + \alpha r_\theta(I_t)\}, \qquad r_\theta(I_t) = \tfrac{1}{2}\tanh g_\theta(I_t). \tag{3}$$

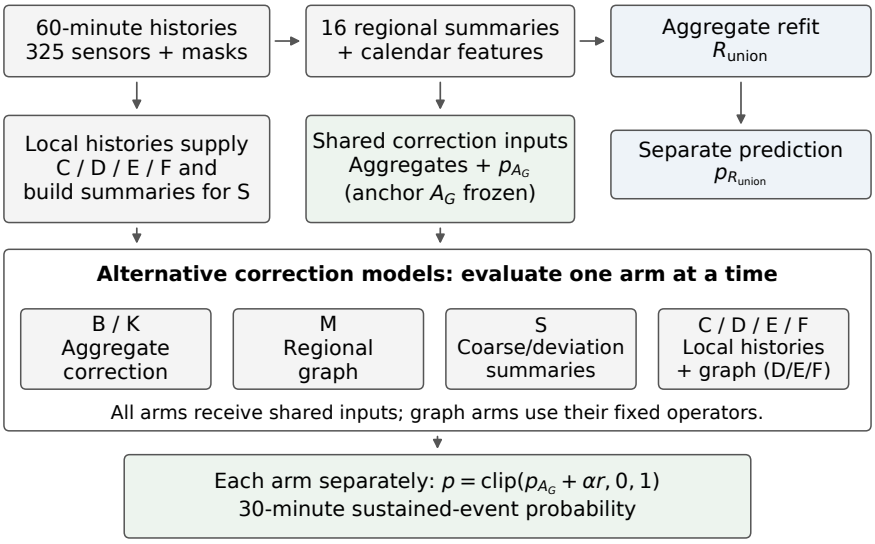


Fig. 1. Prediction paths evaluated as separate models. R_{union} is an aggregate refit with its own output. All corrections use the frozen anchor A_G. B and K use aggregate features, M adds regional messages, S adds graph summaries, and C, D, E and F use local histories. Computing the initial summaries and the inputs to S requires readings from every sensor.

Here, I_t contains the correction inputs, g_θ is the learned network, and α controls correction strength. The residual head was initialized to zero and trained using all-time Brier loss, a penalty λr_θ^2, and weight decay. A zero correction reproduces the anchor exactly, although fitting does not guarantee improved generalization. The global scale $\alpha \in \{0, .25, .5, 1\}$ was selected in a separate training block and applied before clipping. All fitted parameters and selected settings remain unchanged during final evaluation.

4.2 Six Representations

The six representations are local histories (C), distance graph (D), regional graph (M), multiscale graph summaries (S), within-region and cross-region graph (E), and dependency graph (F). Table 1 ~~summarizes the models~~also lists the aggregate controls. C encodes each sensor's history in temporal order using a shared MLP. A separate branch processes that sensor's own encoding before pooling with observation masks within each region. D replaces this self message with a normalized weighted average of information from neighboring sensors. Its graph operator is a distance-similarity matrix whose rows sum to one. Each row specifies the sources contributing to a receiving sensor. The weights represent proximity and do not establish directed road flow or causal influence.

Table 1. Model inputs and processing. Every correction also receives the aggregate vector and frozen anchor probability. Parameter counts exclude the shared anchor.

Model	Extra input	Representation	Parameters
A_G, R_{union}	None	Aggregate tree predictors	Trees
A_{mono}	None	Monotone anchor map	2
B	None	Aggregate correction	38,209
K	None	Larger aggregate correction	50,129
C	Full local histories	Shared temporal encoder with self messages	49,553
D	Full local histories	Sensor messages using distance similarity	49,553
M	None	Messages among regional summaries	49,937
S	Summaries from local histories	Coarse and deviation summaries at 1, 2, 4 hops	50,321
E	Full local histories	Separate messages within and across regions	49,809
F	Full local histories	Sensor messages using a fixed dependency graph	49,553

M encodes regional summary histories and exchanges messages through the fixed regional operator $Q = RWL$, without additional local measurements. S supplies graph-dependent regional summaries at one, two and four hops. E ~~separates messages from neighbors within the same region and those across regional boundaries~~splits D's distance graph by regional membership, separately aggregates messages within and across regions, then concatenates both with the sensor's own encoding. Both neighbor streams use a common denominator based on all available neighbors, preserving their relative contributions. E_internal removes the cross-region stream while retaining this denominator and the same architecture.

F uses D's message architecture with a sparse dependency graph estimated from training data. Calendar effects and regional common components fitted during earlier training are first removed from sensor histories. Neighbors are selected using absolute correlations between the residuals, with the number of nonself neighbors in each row matched to D. Both positive and negative correlations therefore produce positive averaging weights. Matching row degrees does not preserve every indegree. The graph remains fixed during predictor training, and its associations do not identify causal road directions. Spatial relationships thus enter through sensor messages (D, E, F), regional messages (M), or graph summaries (S), before the correction head in Eq. 3.

Temporal encoders and message branches have width 16. The aggregate and final hidden layers generally have width 32, while K uses aggregate width 42. Local and graph models use a common regional output interface where feasible. They still differ in input dimension, channel count, normalization and the functions they can represent. Similar parameter counts help control model capacity but do not ensure equally easy optimization. Accordingly, B versus C assesses

the performance of the fitted representations. It does not measure the value of local information independently of the learner used to process it.

4.3 Coarse State and Graph-Dependent Detail

Let $L \in \mathbb{R}^{325\times16}$ assign each sensor its regional value, and let $R \in \mathbb{R}^{16\times325}$ average sensor values within each nonempty region. Then $RL = I$. For a fully observed sensor vector x, define

$$z = Rx, \quad u = x - Lz, \quad Q = RWL, \quad s_h = RW^h u. \tag{4}$$

Here, z contains regional means, u contains deviations from those means, and Q is the regional operator induced by the sensor graph W. The term s_h captures the contribution of local deviations after h graph steps. Thus

$$RWx = Qz + s_1, \qquad RW^h x = (RW^h L)z + s_h. \tag{5}$$

S supplies the regional and deviation contributions separately at one, two and four hops. Repeated sparse matrix products compute them without forming dense powers of W. In general, $RW^hL \neq Q^h$ because sensor deviations omitted by regional averaging can influence later regional states.

The deviation contribution s_1 vanishes for every x exactly when $RW(I - LR) = 0$, equivalently $RW = QR$. This closure condition means that regional means alone determine the next regional means under W. For directed operators, $WL = LQ$ alone does not imply this condition [8]. These identities explain the construction of S but do not establish that its features are sufficient to predict the nonlinear congestion event.

When inputs are missing, S computes regional means from observed values and sets unavailable deviations to zero. This is equivalent to filling missing entries with the observed regional mean, or zero for an empty region. Equation 5 then applies to the completed feature vector. Availability fractions accompany these features. This fixed-operator construction does not directly describe the normalization that depends on observation masks in D, E and F. Missing future labels remain bounded rather than imputed.

4.4 Topology Controls and Attribution

C provides D's matched control using only self messages. E has separate controls using self messages or internal messages only. Two fixed random seeds generate rewired graphs through directed double-edge swaps without using event outcomes. Each graph preserves the nonself edge count, each node's incoming and outgoing degrees, the multiset of weights in each row, and diagonal fallbacks. Rewiring changes neighbor identities, geometry and regional mixing. In particular, cross-region edges increase from 1,130 to 2,228 in each control. Comparisons with these graphs therefore assess the combined effects of these changes, without isolating geometry from regional mixing.

The two fixed rewired graphs are not uniform samples from all graphs satisfying the degree constraints. Each receives the same tuning budget and all three training seeds as its corresponding original graph model. The comparison evaluates predictors trained with alternative graphs, rather than replacing the graph after training.

5 Experimental Design

5.1 Chronology, Frozen Choices and Exposure

The original split contains 108 complete training days through April 19, 36 validation days from April 20 to May 25, and 36 test days from May 26 to June ~~30.~~30, 2017. March 12 is incomplete and excluded from training. Input and outcome windows must remain within their assigned split, with a 90-minute embargo on both sides of each boundary. The released timestamps contain no timezone information, so we do not assign a timezone convention. At each eligible five-minute test origin t, frozen models use frames $t-11,\ldots,t$ to predict the event in $t+1,\ldots,t+6$. The 9,315 scoreable onset origins are evaluated against these subsequent observations without refitting.

Training is divided into six periods with distinct purposes. Anchor fitting ends on February 23, and correction fitting uses February 24 to March 16. Early stopping uses March 17 to 24, followed by shrinkage and model-family selection from March 25 to April 1. Calibration uses April 2 to 9, and a final development check uses April 10 to 19. Windows crossing these boundaries are excluded, and each period uses only labels observable by its cutoff. The anchor has 5,173 eligible fitting examples, while the aggregate refit has 7,070. Correction fitting uses the later training examples at stride three, with those same examples included in the refit's training union. This controls for the additional training data available to corrections. The earlier pilot had instead compared an early baseline fitted at stride 12 with corrections fitted on more densely sampled later data.

Each correction family had two predeclared regularization settings, evaluated with one tuning seed. The selected setting was then fitted with three retained seeds. AdamW used learning rate .001, weight decay .0001, batch size 256, a maximum of 60 epochs, and early stopping patience of eight epochs. Training minimized unweighted all-time Brier loss with residual regularization. Model selection used development onset Brier. Event observations received no additional weight. Results from earlier rolling chronological evaluations are retained, including unfavorable findings. Overlapping periods and limited event counts mean that these evaluations are not independent studies.

Validation results are exploratory because all validation days had been examined during earlier development. An earlier pilot found favorable results for onset prediction as a secondary endpoint, motivating its registered primary role in the main graph comparison. The strengthened comparison subsequently provided less favorable evidence for the tested local representations. This development history is retained so that the earlier exploratory finding remains distinct from the later registered analysis.

The test measurements were first accessed after committing the complete protocol and verified implementation. All required checkpoints were available, with their hashes, graph operators, feature transforms, shrinkage settings, calibration maps, alarm thresholds and training seeds fixed in advance. Test outcomes were not used to select fitted models, exclusion rules, thresholds, representations or manuscript illustrations. Test and validation results are reported separately, without pooling losses or event groups.

5.2 Metrics, Uncertainty and Alarms

The primary metric is onset Brier score before calibration: the mean squared probability error $N^{-1}\sum_t (p_t - Y_t)^2$ over N scored onset origins. Brier score over all eligible origins is secondary, and results after frozen calibration are reported separately. Lower scores indicate better probability accuracy. For each model family, we average the losses from its three training seeds at each forecast origin before temporal resampling. Probabilities are not combined into an ensemble. Scores for individual seeds and their descriptive standard deviation are also retained.

The four registered primary comparisons are C minus B, D minus C, E minus D, and E minus B. Negative differences favor the first model. We use 2,000 paired resamples of calendar blocks with random seed 20261007. Blocks of three days form the primary analysis, while blocks of one day assess sensitivity. All origins within a selected day remain together, preserving their pairing across models and training seeds. Contiguous calendar segments are resampled separately to preserve gaps. Bonferroni adjustment for the four comparisons uses percentile intervals with 98.75% marginal coverage, targeting approximately 95% coverage across the comparison family. Pointwise 95% intervals are also reported. These intervals describe temporal uncertainty conditional on the fitted models and do not capture all uncertainty from training or development choices.

B versus R_{union} is a required secondary comparison. Additional comparisons involving aggregate models and topology controls are descriptive and use pointwise intervals. F remains secondary regardless of its observed ranking. The protocol's practical reference scales are an absolute Brier reduction of .0001 and a relative reduction of 1% for local inputs. For graph processing, the corresponding values are .00005 and .5%. Alarm tolerances are a recall loss of two percentage points or an increase of one percentage point in the false positive rate (FPR). These scales guide interpretation and do not define equivalence or replace uncertainty analysis.

Each predictor has a logistic calibration map constrained to have positive slope. It was fitted on separate training predictions generated outside that predictor's fitting data. The calibration period contains only 30 positive onset origins in five groups across four days. An alarm occurs when $p > c$, where the cutoff c was fixed to target 5% FPR during calibration. Predictions equal to the cutoff do not trigger alarms, and no noise is added to break ties. The calibration map remains nondecreasing after endpoint clipping and cannot distinguish predictions tied at zero. Test recall and FPR are measured directly because the

calibration target need not hold on test. Recall is the fraction of positive origins alarmed; FPR is the fraction of nonevent origins alarmed.

Positive forecast origins belong to the same group when successive origins are less than 30 minutes apart. A gap of at least 30 minutes starts a new group. A group is detected if any of its origins triggers an alarm. False alarm groups apply the same rule to alarmed origins without an event. These groups summarize repeated adjacent forecasts but are not verified independent traffic incidents. Loss contributions from event and nonevent origins, reliability within fixed probability bins (mean predicted probability versus observed event frequency), and clipping diagnostics complement the overall Brier scores.

5.3 Computation and Verification

Scientific computations ran on one NVIDIA RTX 6000 Ada Generation GPU using PyTorch 2.8.0, CUDA 12.8 and driver 570.124.06. CPU tasks included orchestration, file processing and rendering. Before test access, replay reproduced the frozen raw and calibrated validation predictions exactly. Test predictions were also reproduced exactly, with metric differences within the declared 10^{-12} tolerance. Reproducibility records retain checkpoint hashes, alignment checks, source revisions, resource usage and replay results. The timing results below use previously recorded measurements.

6 Results

6.1 Test Population and Aggregate Baselines

The test period contains 10,333 eligible forecast times, of which 10,327 can be scored. Six fail the future observation coverage requirement. None has conflicting event bounds. Across the full test population, 1,147 origins have positive labels. The onset subset contains 9,315 origins and 260 positives, giving prevalence 2.79%. These positives form 45 groups across 25 days. Validation contains 8,981 onset origins, 289 positives and 47 groups across 26 days. The two populations are evaluated separately.

Aggregate refitting gives the strongest observed scores among the main model families (Table 2). For R_{union}, raw onset and all-time Brier are 0.01425 and 0.01852. Its calibrated scores are also the lowest, at 0.01286 and 0.01712. Aggregate correction B has raw onset Brier 0.01521.

The observed advantage of refitting remains statistically uncertain. The secondary difference B minus R_{union} is .0009656, with a pointwise interval of [−.0000081, .0025347] using blocks of three days. Because this interval includes zero, it establishes neither superiority nor equivalence. The validation difference was near zero. This is an informative comparator because refitting uses the same combined training examples without adding local histories.

The simple probability map A_{mono} also competes with the residual corrections, achieving raw onset Brier 0.01467. Here, "raw" means the registered model output before the final calibration step. For A_{mono}, that output already incorporates its fitted logistic map.

Table 2. Brier scores and onset alarms using frozen thresholds. Correction entries average scores across training seeds. Validation results are exploratory, and raw onset Brier is primary. Recall and FPR describe test onset alarms.

Model	Validation onset raw	Test onset raw	Test all-time raw	Test onset calibrated	Recall	FPR
A_G	0.01411	0.01602	0.02060	0.01397	93.5%	4.87%
R_{union}	0.01217	0.01425	0.01852	0.01286	96.2%	5.29%
A_{mono}	0.01360	0.01467	0.01920	0.01381	93.5%	4.87%
B	0.01217	0.01521	0.01974	0.01399	92.8%	5.97%
K	0.01316	0.01535	0.01993	0.01412	93.2%	5.84%
C	0.01240	0.01537	0.01985	0.01406	93.2%	6.32%
D	0.01253	0.01515	0.01979	0.01399	92.8%	6.69%
M	0.01317	0.01512	0.01978	0.01381	92.4%	4.94%
S	0.01308	0.01524	0.01983	0.01386	92.8%	5.07%
E	0.01228	0.01504	0.01973	0.01399	92.8%	7.12%
F	0.01214	0.01538	0.01986	0.01418	92.7%	6.39%

6.2 Primary Comparisons of Representations

Adding local histories through correction C increases prediction error relative to aggregate correction B. The onset Brier increase is 1.05%, or +.0001593, with an adjusted interval of [.0000522, .0002315] using blocks of three days. The interval excludes zero, supporting the same unfavorable direction observed on validation. This finding concerns the fitted local representation.

Distance graph D improves on C in test, reversing the validation result. It reduces onset Brier by 1.43%, a difference of -.0002203. The pointwise interval [−.0003640, −.0000063] favors D. However, the adjusted interval [−.0004364, .0000268] includes zero.

Cross-region correction E further reduces onset Brier by .71% relative to D, a difference of -.0001074. Its adjusted interval is [−.0002318, .0000073]. The direction agrees with validation, but the adjusted test interval no longer excludes zero. Relative to aggregate correction B, E reduces ~~onset Brier by~~ raw onset Brier from 0.01521 to 0.01504, a reduction of 1.11%, or -.0001684, reversing its unfavorable validation result. This improvement also remains uncertain, with adjusted interval [−.0002960, .0000245].

The observed graph improvements exceed the protocol's practical Brier scales, but their uncertainty depends on the resampling block length. Adjusted intervals using one day favor D over C and E over B. The primary analysis using three days does not. Results also vary across training seeds. E improves over B in two seeds and worsens in the third, while C is worse in all three. These seeds share the same event history and are not independent temporal replications.

Across all eligible origins, differences between corrections are generally smaller, and model rankings can change (Fig. 2). E has all-time Brier 0.01973, compared with 0.01974 for B and 0.01852 for aggregate refitting. Dependency graph F

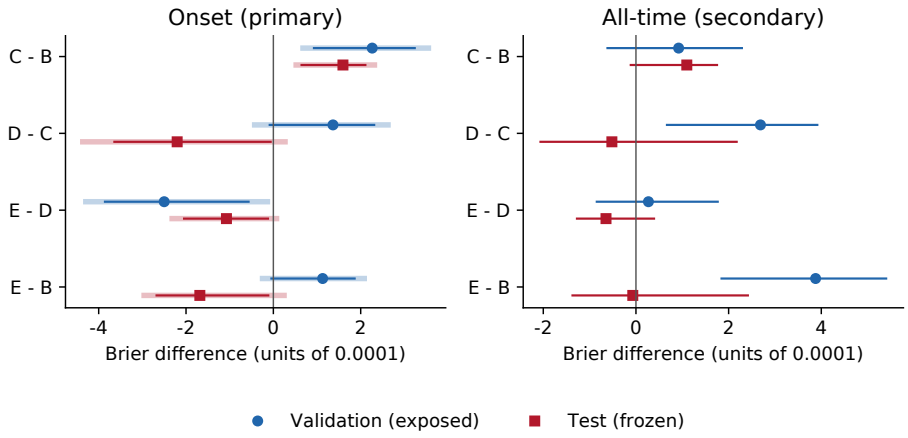


Fig. 2. Paired Brier differences for aggregate correction B, local histories C, distance graph D and cross-region correction E. Thin intervals are pointwise 95% intervals using blocks of three days. Pale thick onset intervals adjust for four comparisons. Values left of zero favor the first model.

had the best validation onset score among corrections but reaches 0.01538 on test and performs worse than B in a secondary pointwise comparison. F remains secondary and does not replace E, which was selected during training.

6.3 Topology and Regional Representation

Cross-region correction E achieves the lowest observed raw onset Brier among the six representations. Regional graph M scores 0.01512 and multiscale summaries S score 0.01524. These results place E first in the observed ranking, although differences between representations remain small relative to temporal uncertainty.

The clearest secondary evidence for E comes from its comparison with trained rewired graphs. E reduces test Brier by .0001886 and .0002492 relative to the two controls. The corresponding pointwise intervals for E minus each control are $[-.0002705, -.0000692]$ and $[-.0004123, -.0000470]$. Both comparisons favor E in all three training seeds. Its comparison with the control using only internal messages is also favorable, while the comparison with E_self remains uncertain.

Distance graph D provides a useful contrast. Its comparison with rewiring seed 05 is uncertain, while seed 04 achieves a lower Brier score by .0000484. The pointwise interval for D minus that control is $[.0000108, .0000854]$. Thus, the original connectivity benefits E relative to both tested rewired alternatives, but does not provide the same consistent advantage for D.

Together, these comparisons show why topology should be evaluated alongside the message architecture that uses it. The favorable results for E are secondary findings and were not consistent across validation and test. Because

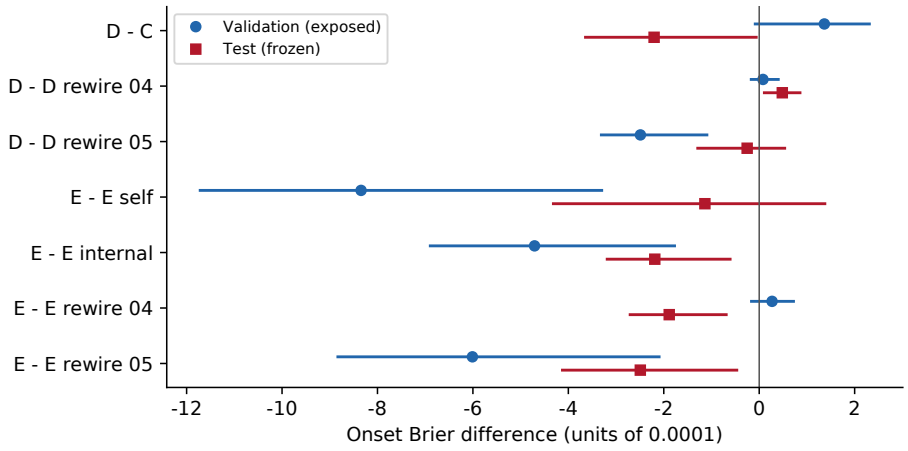


Fig. 3. Distance graph D and cross-region correction E compared with trained topology controls. Pointwise intervals use blocks of three days. Rewiring preserves degrees and row weights while changing regional mixing. D versus C is also a primary comparison. Comparisons with the additional rewired and internal controls are descriptive secondary analyses.

rewiring changes regional mixing as well as neighbor identity, the comparisons assess these aspects of graph organization together.

6.4 Alarms, Calibration and Error Structure

The models detect most positive forecast groups. Aggregate refitting R_{union} achieves recall 96.15% at FPR 5.29% and detects all 45 groups. B, C, D and E each detect an average of 43.33 groups. Group detection requires at least one alarm within a group.

The comparison also reveals a useful distinction between probability accuracy and alarm quality. B and E achieve the same mean recall of 92.82%, but their FPRs are 5.97% and 7.12%, respectively. E's increase exceeds the registered tolerance of one percentage point, showing that its lower raw Brier score does not translate into fewer false alarms.

Frozen calibration improves test onset Brier for every main model while leaving alarm decisions unchanged. Aggregate refitting retains the lowest observed calibrated score, 0.01286, and B ranks ahead of E after calibration. This benefit differs from validation, where calibration increased error.

Error analysis helps explain these results. E's raw Brier gain over B comes from lower loss on origins without an event, despite higher loss on positive origins. Across B, C, D and E, between 84.2% and 86.6% of raw onset predictions equal zero, limiting the distinctions available to monotone calibration. Higher predicted probabilities also tend to overstate observed risk. ~~These diagnostics show how calibration, probability scores and alarm behavior provide complementary~~

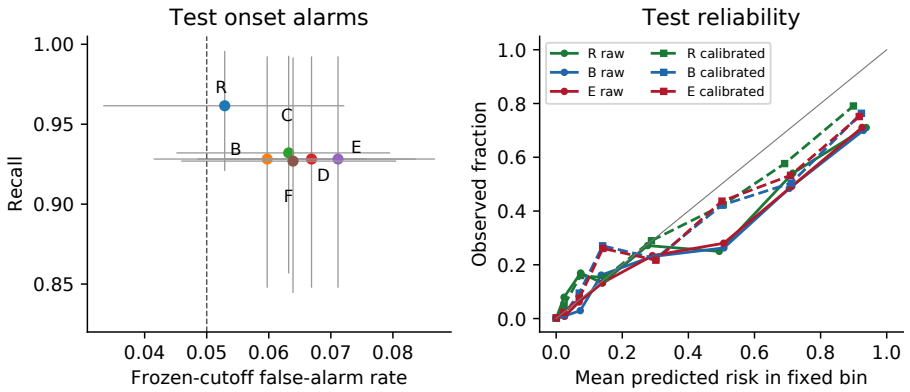


Fig. 4. Test alarm and calibration results. R denotes R_{union}. The vertical reference line marks the training FPR target. Reliability bins pool predictions across seeds and origins without treating them as independent observations. All alarm thresholds remain fixed.

~~information about each representation~~Brier averages errors across all origins, whereas FPR depends on threshold crossings among nonevents. E can therefore lower average nonevent loss while placing more nonevent scores above its frozen cutoff. Limited calibration support and a shift in score distributions between calibration and test could contribute; their separate effects were not established.

6.5 Measured Costs

Table 3 compares inference times for a common validation workload of 128 forecast origins. The measurements include history parsing with a warm cache, data transfers, feature preparation, prediction and output transfer. Every correction includes inference from the shared anchor A_G. Model restoration and graph preparation are recorded separately as setup costs. Future label construction is outside the timed workload.

Each mean summarizes three repeats using the first registered training seed. The measurements provide a consistent basis for comparing computational costs, although overlapping timings across repeats do not support a clear latency ranking.

Payload accounting describes the numerical inputs associated with each representation. Using float32 values, the aggregate vector occupies 4,632 bytes. Full local histories add 15,600 bytes, while the three regional deviation summary streams use 2,304 bytes. These counts cover feature values before masks, sensor identities and graph metadata. They characterize representation size rather than measured communication or memory savings. Both the initial regional summaries and the graph summaries are computed from readings across all sensors.

Table 3. Mean measured inference time for 128 forecast origins, in seconds. Each entry summarizes three repeats. Correction times include the shared anchor.

Aggregate model	Mean seconds	Representation	Mean seconds
A_G	0.2022	C	0.2087
R_{union}	0.2038	D	0.2080
A_{mono}	0.2074	M	0.2052
B	0.2066	S	0.2048
K	0.2061	E	0.2066
		F	0.2068

7 Discussion

The comparison shows how aggregate updating can change the interpretation of graph performance in regional congestion prediction. Aggregate refit R_{union} combines favorable observed Brier scores with strong alarm performance using aggregate inputs. Its paired comparison with B remains uncertain, but its competitiveness establishes the importance of evaluating richer representations against an updated aggregate model. Matching access to training examples makes this comparison more informative by accounting for gains that could arise from additional labels. Refitting re-estimates the full tree ensemble on the combined fitting examples, allowing the aggregate predictor itself to adapt; C instead learns a regularized, bounded correction of a frozen anchor.

The results also distinguish access to individual measurements from their effective use by a predictor. Local correction C performs worse than aggregate correction B on both validation and test. This finding concerns the tested representation and learning procedure, rather than the intrinsic information in local histories. Regional extrema, variation and severe fractions may already summarize patterns relevant to the regional event. ~~The comparison therefore motivates examining what additional predictive structure a local encoder captures beyond these summaries~~Redundant local inputs and the more complex encoder could increase estimation or optimization difficulty under the available training budget; the comparison does not isolate these explanations.

Cross-region correction E provides the clearest secondary evidence for the value of the tested graph organization. It improves on both trained rewired controls in test, with the same direction across all three training seeds. These comparisons assess neighbor identity and regional mixing together. D's different response to rewiring further motivates evaluating graph structure jointly with the architecture that processes it. The resulting evidence concerns particular combinations of connectivity and message processing, while the primary graph improvements remain statistically uncertain.

Alarm evaluation adds a practical dimension to these representation comparisons. E's lower raw Brier score relative to B accompanies a higher false alarm rate without increased recall. Its score improvement comes from lower loss on origins without an event, illustrating how overall probability accuracy can improve

without better event detection. Frozen calibration improves test Brier across the main models and changes their relative ranking while preserving alarm decisions. Probability scores, calibration and alarm rates therefore reveal different aspects of forecast quality.

For regional traffic monitoring, the study supports a structured approach to model selection. Competitive aggregate updating provides the reference, trained topology controls clarify the contribution of connectivity, and alarm evaluation determines how score differences translate into warnings. This combination makes the empirical comparison useful even when a more detailed representation does not consistently outperform the aggregate alternatives.

8 Conclusion

This study provides a controlled comparison of six representations for regional congestion onset prediction. Matched access to training labels, frozen chronological evaluation and trained topology controls clarify the contributions of aggregate updating, local histories and graph processing. Aggregate updating provides a competitive baseline, while the tested local correction does not improve prediction. Correction E outperforms both trained rewired controls in secondary test comparisons, providing evidence in favor of its graph organization. The primary graph improvements remain uncertain, and E raises false alarms without improving recall. These findings offer a reproducible basis for model selection by showing how probability accuracy, topology comparisons and alarm behavior together reveal the practical value of a representation.

AI Assistance

An AI coding and writing assistant supported implementation, literature checking, and manuscript grammar check.

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A Per-Seed Onset Results

Table 4. All frozen test arms. Corrections retain seeds 20261001, 20261002 and 20261003 in order. Null suffixes 04/05 identify graph seeds 20261004/20261005. SD is descriptive with denominator three, not a confidence interval.

Arm	Seed 01	Seed 02	Seed 03	SD
A_G	0.016018	–	–	0.000000
R_{union}	0.014247	–	–	0.000000
A_{mono}	0.014674	–	–	0.000000
B	0.015521	0.015031	0.015085	0.000220
K	0.015741	0.015297	0.015027	0.000294
C	0.015715	0.015171	0.015229	0.000244
D	0.015312	0.014845	0.015296	0.000217
M	0.015341	0.014823	0.015185	0.000217
S	0.015423	0.014899	0.015387	0.000239
E	0.015057	0.014757	0.015317	0.000229
F	0.015668	0.014971	0.015494	0.000296
E_internal	0.015683	0.014834	0.015271	0.000347
D rew. 04	0.015222	0.014739	0.015348	0.000263
D rew. 05	0.015480	0.014859	0.015191	0.000253
E_self	0.015299	0.014832	0.015342	0.000231
E rew. 04	0.015328	0.014803	0.015566	0.000319
E rew. 05	0.015662	0.014769	0.015448	0.000381